

CSE 310: Data Structures and Algorithms (2024 Summer C)



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CSE 310: Data Structures and Algorithms

Course and Faculty Information

Course Description: Advanced data structures and algorithms, including stacks, queues, trees, and graphs. Searching for graphs, hashing, sorting

Credits: 3

Prerequisites: Comp Systems Engineering BSE, Comp Science BS, Informatics BS, Engineering Mgmt BSE or Computational Math Sci BS; CSE 220 or 240 with C or better; MAT 243 with C or better (CMS students may substitute MAT 300) or CSE Graduate student

Instructor: Dr. Janaka Balasooriya

Contact Info: Email: janakab@asu.edu

Office Hours: Monday, Wednesdays from 6.30 - 7.30 pm (Arizona time) or by appointment, <https://asu.zoom.us/j/3342417992> 

[\(https://asu.zoom.us/j/3342417992\)](https://asu.zoom.us/j/3342417992)

Instructor Live Session Mondays from 6.30 - 7.00 pm

<https://asu.zoom.us/j/3342417992>  <https://asu.zoom.us/j/3342417992>): The

Objective of the live session is to discuss each weekly module objectives, key

concepts, and solutions to some of the quiz, homework, and practice questions. We highly encourage you all to join the live session. The session will be recorded and posted so that you can review them later if you can't attend.

Graduate Instructional Assistant:

Please reach out to Mohsen if you have any recitation-related questions:

Name/email: Sheibanian, msheiban@asu.edu

Recitation sessions: Wed 3-4 PM ([ZOOM Link](https://asu.zoom.us/j/84053502830)  [\(<https://asu.zoom.us/j/84053502830>\)](https://asu.zoom.us/j/84053502830)) ---> the first recitation will be on Friday May 17, 3-4 pm

Office hours: Tue / Thr 5 -6 pm ([ZOOM Link](https://asu.zoom.us/my/mohsen?omn=89539225383)  [\(<https://asu.zoom.us/my/mohsen?omn=89539225383>\)](https://asu.zoom.us/my/mohsen?omn=89539225383))

Undergraduate TAs (Please note that undergraduate TAs do not participate in grading, please do not contact them related to grading questions) :

Name/email: Andrew Houghton, ahought1@asu.edu

Office hours: Wednesday at 6:00PM and Saturday at 10:00AM

Zoom Link:

[https://asu.zoom.us/j/5140229011?
pwd=ekFBN0F1WG82V00weW9mRWx4WEJOUT09](https://asu.zoom.us/j/5140229011?pwd=ekFBN0F1WG82V00weW9mRWx4WEJOUT09) 
[\(\[https://asu.zoom.us/j/5140229011?
pwd=ekFBN0F1WG82V00weW9mRWx4WEJOUT09\]\(https://asu.zoom.us/j/5140229011?pwd=ekFBN0F1WG82V00weW9mRWx4WEJOUT09\)\)](https://asu.zoom.us/j/5140229011?pwd=ekFBN0F1WG82V00weW9mRWx4WEJOUT09)

Password: cse310

Name/email: Adrian Will, atwill@asu.edu

Office hours: Monday 3:30 - 5 pm, Tuesday 5 pm - 6.30 pm

Zoom Link: <https://asu.zoom.us/j/9623718183> 
(<https://asu.zoom.us/j/9623718183>)

Name/email: Rui Heng Foo, rfoo1@asu.edu

Office hours: Wednesday and Thursday, 8-9.30 pm

Zoom Link: <https://asu.zoom.us/j/81634291488> 
(<https://asu.zoom.us/j/81634291488>)

Name/email: Aditya Jarodiya, ajarodiy@asu.edu

Office hours: Wednesday 6:30 - 8:00 PM, Friday 6:30 - 8:00 PM

Zoom Link: <https://asu.zoom.us/j/84280864127> 
(<https://asu.zoom.us/j/84280864127>)

 (<https://asu.zoom.us/j/3488829719>)

NOTE:

Also, the contents of this course, including lectures and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling or distributing course content or notes taken during the conduct of the course. Any recording of class sessions is authorized only for the use of students enrolled in this course during their enrollment in this course. Recordings and excerpts of recordings may not be distributed to others.

Course Learning Outcomes

At the completion of this course, students will be able to:

1. Students who complete this course can define data structures (types) such as heaps, balanced trees, hash tables.
2. explain how to use a specific data structure in modeling a given problem (e.g. I can explain how to model a dictionary using balanced trees).
3. identify, construct, and clearly define a data structure that is useful for modeling a given problem.

4. state some fundamental algorithms such as merge sort, topological sort, Prim's and Kruskal's algorithm, and algorithmic techniques such as dynamic programming and greedy algorithms.
5. use a specific algorithmic technique in solving a given problem (e.g. I can write a dynamic program that solves a shortest-path problem).
6. design an algorithm to solve a given problem.
7. define the notions of worst-case/best-case/average-case running times of algorithms.
8. analyze and compare different asymptotic running times of algorithms.
9. analyze a given algorithm and determine its asymptotic running time.
10. combine fundamental data structures and algorithmic techniques in building a complete algorithmic solution to a given problem.
11. create several algorithmic solutions to a given problem and choose the best one among them according to given requirements on time and space complexity.

Major Topics Covered

- Asymptotic Notation
- Worst-case Analysis
- Sorting, Lower Bound for Sorting, Heaps, and Heap sort
- Median, Selection Problems: More on Divide-and-Conquer Algorithms
- Hashing
- Binary Search Trees, Red-Black Trees
- Disjoint Sets (union-find): Introduction to Amortized Analysis
- Longest Common Subsequence: Introduction to Dynamic Programming
- Graph Algorithms (Depth-First Search, Breadth-First Search, Topological Sort, MST: Greedy
- Algorithms, Shortest Paths

Tentative Schedule (Summer 2024)

Week	Topics	Assignment/Exams

1 (05/16-05/19)	Introduction Module: Syllabus, Asymptotic Notation (sec 3.1), Insertion Sort (sec 2.1), More on asymptotic Notation (sec 3.1).	
2 (05/20-05/26)	Module 1: Merge Sort (sec 2.3), Recurrences (chap4), Quick Sort (chap7).	HW1
3 (05/27 – 06/02)	Module 2: Heaps (sec 6.1,6.2), Heap Sort (sec 6.3,6.4), Priority Queues (sec 6.5), Sorting Lower Bound (sec 8.1,8.2), Selection (chap9),	
4 (06/03 – 06/09)	Module 3: Binary Search Trees (chap12). Midterm I - Online	HW2 Midterm I (Available on Monday 10th from 5pm – 11:59 pm)
5 (06/10 – 06/16)	Module 4: Hash Tables (chap11), Red-Black Trees (chap13),	HW3
6 (06/17 – 06/23)	Module 5:	

	Graph (sec 22.1), Breadth-First Search (sec 22.2), Depth First Search (sec 22.3), Topological Sort (sec 22.4), Disjoint Sets (chap21).	
7 (6/24 - 6/30)	Module 6: Minimum Spanning Trees (chap23), Shortest Paths (chap24) Shortest Path Algorithms	HW 4 Midterm II (Available on Saturday 29th from 5pm – 11:59 pm)
8 (7/01 - 7/07)	Module 7, Module 8: Dynamic Programming (chap 15)	HW5
9 (7/08 - 7/10)	Final Exam (Available on Wednesday July 10th from 5pm – 11:59 pm, Arizona Time)	

Textbooks

- Cormen, T. H., Leiserson, C.E., Rivest, R.L., and Stein, C., Introduction to Algorithms, The MIT Press, Cambridge Massachusetts, 3rd ed., 2009. or 4th Edition. **The textbook is absolutely needed.**

Course Access

Your ASU courses can be accessed by both my.asu.edu (<http://my.asu.edu>) and myasucourses.asu.edu (<http://myasucourses.asu.edu>); bookmark both in the event that one site is down.

Class Format/Attendance

All the lectures are pre-recorded and available to students each week. However, during the **live session on Monday from 7.30 - 8.00 pm** Instructor will discuss key concepts and conduct problem solving activities. This zoom session will be recorded. However, students are strongly advised to attend and participate. Thursday is the recitation day and the recitation and the TA will conduct the recitation.

Recitation attendance is expected (not mandatory). Recitations will be on **Wednesday 3-4 pm**. Recitations will be conducted by the TA. Recitations will be recorded. However, attending the recitations is highly recommended and recitation TA will take attendance even if points are not given for attendance

Computer Requirements

You need to have a computer that is capable of compiling and running C++ programs. Software that we use in the class is Visual Studio 2015 or above. BYENG 214 lab computers have Visual Studio installed. Mac users can use VS Code or XCode. We use g++ compiler to test programs

Student Success

To be successful:

- check the course daily
- read announcements
- read and respond to course email messages as needed
- complete assignments by the due dates specified
- communicate regularly with your instructor and peers
- create a study and/or assignment schedule to stay on track

Grading

Assessment & Grading

Assessment	Percentage
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Exams (including the Final) :Midterm I - 10%, Midterm II - 15% Final Exam - 20%	45%
Recitation - 15% Quizzes - 10%	25%
Homework	30%
lecture review activities	5% Extra Credit
Total	105%

Exams: There will be absolutely no make-up exams. Your picture ID needs to be shown during the exams. Exams will be proctored using Lockdown browser with WebCam

Midterm exam dates: See the schedule.

Quizzes : Similar to exams, there will be absolutely no make-up quizzes. Your picture ID needs to be shown during the quizzes. Quizzes will be proctored using Lockdown browser with WebCam. We will have 4-5 quizzes during the semester.

Assignments: Assignments will include programming exercises. There will be about 5 assignments. **No late assignment will be accepted unless satisfying the following late/missed assignments policy**

Late or Missed Assignments

If you need to miss a quiz/test/assignment, you must have a legitimate university approved excuse, such as a severe injury, illness, or participation in a legal proceeding that require your presence, etc. You must contact the instructor and provide satisfactory evidence prior to or within two days after the absence. A missed quiz/test/assignment without permission or supporting documents is a zero.

Note: *Please note that we will not be able accept late submissions or reschedule exams due to personal trips or vacationing.*

Notify the instructor BEFORE an assignment is due if an urgent situation arises and you are unable to submit the assignment on time.

Follow the appropriate University policies to request an [accommodation for religious practices](http://www.asu.edu/aad/manuals/acd/acd304-04.html) [\(http://www.asu.edu/aad/manuals/acd/acd304-04.html\)](http://www.asu.edu/aad/manuals/acd/acd304-04.html) or to accommodate a missed assignment [due to University-sanctioned activities](http://www.asu.edu/aad/manuals/acd/acd304-02.html) [\(http://www.asu.edu/aad/manuals/acd/acd304-02.html\)](http://www.asu.edu/aad/manuals/acd/acd304-02.html).

Final Grade Percentage

A+ $\geq 97\%$

A $\geq 90\%$ and $< 97\%$

B+ $\geq 87\%$ and $< 90\%$

B $\geq 80\%$ and $< 87\%$

C+ $\geq 77\%$ and $< 80\%$

C $\geq 70\%$ and $< 77\%$

D $\geq 60\%$ and $< 70\%$

E $< 60\%$

Fulton Schools of Engineering (FSE)

Academic Integrity Policy

Students in this class must adhere to ASU's academic integrity policy, which can be found at <https://provost.asu.edu/academic-integrity/policy> [\(https://provost.asu.edu/academic-integrity/policy\)](https://provost.asu.edu/academic-integrity/policy)). Students must review this policy and become familiar with each of the areas in which academic dishonesty can occur. All academic integrity violations will be reported to the Fulton Schools of Engineering Academic Integrity Office. The Academic Integrity Office (AIO) maintains a record of all violations and has access to academic integrity violations committed in all other ASU colleges/schools. Course content, including lectures, are copyrighted materials. In addition to ASU's academic integrity policy, students may not share outside the class, upload, sell or distribute course

content or notes taken during the conduct of the course (see [ACD 304–06 \(https://www.asu.edu/aad/manuals/acd/acd304-06.html\)](https://www.asu.edu/aad/manuals/acd/acd304-06.html), “Commercial Note Taking Services” for more information).

students must refrain from uploading to any course shell, discussion board, or website used by the course instructor or other course forum, material that is not the student's original work, unless the student first complies with all applicable copyright laws; faculty members reserve the right to delete materials on the grounds of suspected copyright infringement

Collaboration for Assignments

Unless otherwise instructed, you can discuss the assignments with other students and exchange ideas about how to solve them. However, there is a thin line between collaboration and plagiarizing the work of others, i.e. cheating. In order that you do not cross that line, you are required to compose your own unique solution to each of the problems. You cannot use any code or solution written by any of your classmates.

Some obvious acts of cheating are:

- * Turning in work/code done by someone else.
- * Copying work/code done by someone else.
- * Writing one code/solution together with someone else (it should be individual work).

All instances of cheating will be handled by the Dean's office according to the *FSE Academic Integrity Policy listed above*.

Title IX

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic

support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <http://sexualviolenceprevention.asu.edu/faqs/students>.